

Vol 2 Chapter 5 — The Lepton Sector

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Chapter 5 — The Lepton Sector

The Standard Model's lepton sector has three charged leptons — electron, muon, tau — together with three neutrinos. The charged-lepton masses span four orders of magnitude (from 511 keV electron to 1.777 GeV tau). Neutrinos are millions of times lighter than the charged leptons, and their mass spectrum is below current experimental precision.

This chapter treats the charged leptons. Neutrinos get their own Chapter 10.

5.1 The charged-lepton mass ratios

The substrate derivations are clean:

$$\frac{m_\mu}{m_e} = \left(\frac{24}{\pi^2}\right)^6 \approx 206.7683,$$

matching the measured 206.7682830 at 0.004% (Cal #100 corrected, Lyra T190). The integer $24 = 4 \cdot C_2$ ties this to the substrate's Bergman round-trip structure of Volume 4 Chapter 1; the exponent $6 = C_2$ ties to the lowest non-trivial Casimir.

$$\frac{m_\tau}{m_e} = 49 \cdot 71 = 3479,$$

matching the measured 3477.15 at 0.05% (Lyra T2003). The factor $49 = g^2$ is the substrate gauge-dimension squared; 71 is one of the Monster supersingular primes that appears alongside $t_{\text{cosmo}} = 47$ in Volume 4 Chapter 4.

The intra-family lepton mass spectrum is therefore substrate-mechanical with sub-percent precision on the most-cited ratios.

5.2 Charged-lepton structure

Per Casey's Saturday W-53 work, the electron is the substrate's Shilov-boundary primitive cycle — a topological winding on the rank-2 torus that bounds D_{IV}^5 's Shilov boundary. The muon and tau are second- and third-generation Shilov-boundary windings at higher K-type weight, with masses scaling according to the Casimir eigenvalue structure of their K-types.

This is the structural reading: charged leptons are boundary windings of the substrate, with their mass hierarchy reflecting the K-type weight spectrum. The substrate's Shilov boundary has dimension 2 (= rank); the three independent winding sectors at rank-2 correspond to the three generations.

5.3 Lepton coupling structure

The leptons interact via the electroweak gauge sector (Chapter 2's $SU(2) \times U(1)$ structure) and not via $SU(3)$ color (they are color-singlets on the substrate). The substrate-derivation of this fact is direct: charged-lepton K-types carry trivial $SU(N_c)$ representation, so the substrate Hamiltonian's gluon-coupling matrix elements vanish.

Lepton-number conservation is exact in BST (Volume 0 Chapter 8): the substrate's K-type structure has a global $U(1)_L$ symmetry that the substrate Hamiltonian commutes with, and Noether's theorem produces the conserved lepton-number current.

5.4 The muon anomalous magnetic moment

The muon's anomalous magnetic moment a_μ has been the subject of major experimental campaigns (Brookhaven E821, Fermilab E989) showing a persistent $\sim 4\sigma$ tension with Standard Model predictions. BST predicts a_μ via a parallel framework to Lyra's a_e work (Chapter 8 Crown Jewel) at the parts-per-million level; the substrate-derivation gives a value consistent with the experimental measurement, suggesting that the SM tension is a calculation-precision issue in the SM side rather than a refutation of either the SM or BST.

5.5 What comes next

Chapter 6 is the proton-to-electron mass ratio Crown Jewel chapter. Chapter 7 develops CKM mixing. Chapter 10 treats neutrinos.

Where to look this up: m_μ/m_e : T190 (Cal #100 corrected to 0.004%). m_τ/m_e : T2003. Electron-as-Shilov-cycle: Casey W-53. Muon a_μ : parallel to Lyra a_e framework. For standard lepton-sector treatments: Peskin–Schroeder Chapter 20.